

Agamjot Nijjar

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EDUCATION

UNIVERSITY OF TORONTO MISSISSAUGA | Mississauga, ON

2023 — 2028

Bachelor of Science, Geospatial Data Science

Coursework: Geographic Information Systems, Spatial Data Analysis, Computer Programming, Python for Data Science

WORK EXPERIENCE

INVENTORY & DISTRIBUTION SYSTEMS ASSOCIATE — TJX Canada

May 2025 — August 2025

- Tasked with processing **200+** incoming and outgoing data center shipments per shift; maintained **99%** inventory accuracy by implementing a double-verification scanning workflow that eliminated mislabeling errors.
- Operated **industrial cutting machines** to process and cut tickets to specification, and performed hands-on **troubleshooting** of mechanical faults and calibration issues to minimize downtime and maintain uninterrupted production flow.
- Maintained detailed logs of hardware movements and flagged **12+** discrepancies over the term, enabling supervisors to resolve inventory conflicts before they escalated to audits.

COMPUTER SCIENCE INSTRUCTOR — Megamind Learning Centre

June 2024 — August 2024

- Delivered **daily lessons** in computer science and mathematics to groups of **15–20** students aged 8–14; developed visual, activity-based exercises that improved end-of-week assessment scores by **30%**.
- Taught a structured **troubleshooting** approach for common software and hardware issues, reducing instructor intervention requests by **25%** over the 8-week program.
- Communicated weekly progress updates to **parents and coordinators**, maintaining consistently positive feedback on student engagement and learning outcomes throughout the term.

PROJECTS

PEEL REGION HOSPITAL COVERAGE MAPPER

Project 1

- Developed a **GIS-based population catchment analysis tool** in Python and ArcGIS Pro that mapped hospital service areas across Peel Region, Ontario, processing **1,746 dissemination area boundaries** for a population of **1.45 million** residents.
- Engineered a **coordinate system projection pipeline** using arcpy's PointGeometry and projectAs functions to convert hospital GPS coordinates from **WGS84** to **NAD83 Statistics Canada Lambert** projection, enabling accurate Euclidean distance calculations across all 1,746 spatial units.
- Processed and filtered a **740MB** Statistics Canada 2021 Census CSV using **pandas chunk-based reading** (100,000 rows per iteration) to extract Peel Region demographic data, joining population counts to dissemination area boundary shapefiles via **DAUID key matching**.
- Automated end-to-end **geodatabase population** using arcpy UpdateCursor, writing nearest-hospital assignments directly into ArcGIS feature class attributes and visualizing catchment zones as a **choropleth map** symbolized by unique hospital service areas in ArcGIS Pro.

FLIGHT ROUTE TRACKER

Project 2

- Built a **geospatial mapping application** in Python using **Folium** and **Pandas** that ingests **5,000+** airline route records and renders them as an interactive HTML world map, engineering a **color-coding algorithm** that assigns a unique, consistent color to each route across all airports it services by processing latitude/longitude coordinate pairs into precise polyline flight paths.
- Cleaned and normalized inconsistent coordinate formats across **major US and Canadian airports** to achieve a **100%** render success rate with zero missing paths, then exported the result as a **standalone HTML file** deployable in any browser, demonstrating end-to-end skills in data ingestion, spatial processing, and interactive web-based visualization.

TECHNICAL SKILLS

Programming & Data: Python, C++, Java, R, Folium, HTML, Data Visualization, Microcontroller Programming

Tools & Embedded: ArcGIS Pro, GitHub, VS Code, Arduino, Circuit Design, R Studio, AutoCAD, Google Sketchup

Certifications: Data Science 101 by IBM, Python 101 for Data Science by IBM **Languages:** English, Punjabi, Hindi